

Closing the Verification Gap: Non-Intrusive Auditing of Lighting Efficiency in Commercial Buildings

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Abstract. A significant *verification gap* exists in sustainable architecture, where the designed energy efficiency of buildings is rarely achieved in practice, leading to substantial energy and financial waste. This gap persists because the prohibitive cost of hardware sub-metering prevents direct performance auditing at scale. This paper presents a fully unsupervised, two-stage deep learning framework that enables scalable auditing of lighting efficiency using only aggregate electricity meter data. Our methodology is validated on the public ASHRAE Great Energy Predictor III dataset, analyzing nearly 500,000 daily load profiles from 1,448 buildings across 16 climate zones.

In the first stage, a 1D Convolutional Autoencoder learns a latent representation of daily energy profiles, enabling robust clustering of operational patterns. This approach yields a Silhouette Score of 0.37, significantly outperforming baseline clustering on raw data (0.35) and revealing three distinct, interpretable lighting control regimes. In the second stage, an N-BEATS model performs blind source separation, decomposing the aggregate signal into a baseline trend and the isolated lighting load. Our model achieves superior reconstruction accuracy over Gradient Boosting baselines while delivering physically interpretable decompositions that classical STL cannot provide. Extensive experiments confirm the model’s robustness to noise and its ability to generalize across different climates. A practical case study on an educational facility illustrates how the framework can identify specific, actionable energy-saving opportunities without any hardware retrofits. This work provides a key lesson in applying unsupervised deep learning to real-world problems where ground truth is absent, offering a transparent and reproducible methodology for the applied data science community.

Keywords: Applied Data Science, Building Energy Management, Non-Intrusive Load Monitoring, Energy Disaggregation, Deep Learning, Sustainable Design

1 Introduction

Commercial buildings account for a substantial portion of global energy consumption, with lighting alone representing 15% to 20% of total electricity use [5]. Despite stringent energy performance standards mandated during the design phase, a pervasive “verification gap” remains—a well-documented discrepancy between predicted efficiency and actual operational consumption [2][3]. A primary driver of this gap is suboptimal operational control, particularly in systems highly sensitive to occupant behavior [4]. Static schedules often fail to adapt to dynamic occupancy patterns, leading to substantial energy waste, such as illuminating empty floors during evenings or weekends.

Addressing this inefficiency requires granular, continuous auditing of specific end-use loads. Unfortunately, direct measurement via hardware sub-metering (installing dedicated physical meters on every lighting circuit) is prohibitively expensive, disruptive, and logistically complex for the vast majority of existing commercial building stock [7]. Consequently, facility managers are left with whole-building aggregate smart meter data, which provides a macro-level view of consumption but obscures the specific operational patterns of individual systems like lighting.

To bridge this gap, Non-Intrusive Load Monitoring (NILM)—also known as energy disaggregation—has emerged as a powerful computational alternative to hardware sub-metering [6]. NILM frames the extraction of individual appliance loads from an aggregate power signal as a blind source separation problem. While high-frequency NILM (sampling at kHz to MHz rates) has shown success in identifying transient electrical signatures of specific devices [8], commercial smart meters typically report data at much lower frequencies, such as hourly intervals. At this resolution, transient electrical signatures are lost, and disaggregation must rely entirely on macroscopic temporal patterns.

Recent advancements in deep learning have significantly advanced the state-of-the-art in low-frequency NILM. Architectures such as Convolutional Neural Networks (CNNs) [12], Recurrent Neural Networks (RNNs), and Transformers [13] have demonstrated the ability to extract complex temporal dependencies from aggregate load profiles. However, many of these approaches are heavily supervised, requiring large datasets of paired aggregate and sub-metered ground-truth data for training. In the context of commercial buildings, such ground-truth data is exceedingly rare, rendering supervised approaches impractical for widespread deployment.

This paper proposes a fully unsupervised, two-stage deep learning framework designed to audit lighting efficiency in commercial buildings using only standard, hourly aggregate electricity meter data. Our approach does not require any historical sub-metered data or hardware retrofits. Instead, it leverages the inherent structural differences between continuous base loads (e.g., HVAC, servers) and discrete, occupancy-driven loads (e.g., lighting) to perform the disaggregation.

The core contributions of this work are as follows:

1. We introduce an unsupervised feature learning pipeline utilizing a 1D Convolutional Autoencoder (1D-CAE) to project daily aggregate load profiles into

a compact latent space. This enables robust, scalable clustering of building operations into distinct control regimes (e.g., scheduled weekday use versus reactive weekend use).

2. We adapt the N-BEATS (Neural Basis Expansion Analysis for Interpretable Time Series) architecture [15]—originally designed for forecasting—as a powerful unsupervised signal decomposition tool. Specialized trend and seasonality blocks recover interpretable high-frequency components consistent with lighting behavior, separating them from the low-frequency HVAC and base load components.
3. We rigorously validate our framework on the public ASHRAE Great Energy Predictor III dataset [1], processing nearly 500,000 daily load profiles from 1,448 buildings across 16 climate zones. Our evaluation includes comprehensive baselines, a cross-climate generalization test, and a sensitivity analysis demonstrating the model’s robustness to input noise.
4. We provide a practical, end-to-end case study on an educational facility, illustrating how the extracted lighting profiles can be directly used by facility managers to identify actionable energy-saving opportunities, thereby directly addressing the verification gap.

The remainder of this paper is structured as follows. Section 2 reviews related work in building energy performance, NILM, and deep learning for time series. Section 3 details the proposed two-stage methodology. Section 4 presents the experimental setup, baseline comparisons, and the practical case study. Finally, Section 5 concludes the paper and discusses avenues for future research.

To ensure reproducibility, all code and trained model weights will be released upon acceptance.

2 Related Work

2.1 Building Energy Auditing and the Performance Gap

As introduced earlier, the building energy performance gap—where actual consumption frequently exceeds design predictions by a factor of two to three [4]—is driven largely by occupant behavior and operational control strategies [2]. Addressing this gap requires continuous, data-driven operational auditing. Traditional audits are labor-intensive, relying on manual inspections and the temporary installation of sub-metering equipment. While whole-building smart meters have become ubiquitous, they lack the granularity required to pinpoint inefficiencies in specific subsystems. This limitation has spurred significant interest in computational methods capable of extracting subsystem-level insights from aggregate meter data.

2.2 Non-Intrusive Load Monitoring (NILM)

Non-Intrusive Load Monitoring (NILM), first conceptualized by Hart in the early 1990s [6], aims to disaggregate a total electrical load into its constituent

appliance-level components using only a single measurement point. NILM is fundamentally a single-channel blind source separation problem.

Historically, NILM research has focused heavily on the residential sector, driven by the availability of high-frequency datasets (e.g., REDD [16], UK-DALE [17]) that capture voltage and current waveforms at kHz or MHz sampling rates. At these frequencies, individual appliances exhibit unique, transient electrical signatures (e.g., the specific current draw profile of a refrigerator compressor starting) [7][8].

In contrast, commercial buildings present a significantly more challenging environment for NILM. Commercial smart meters typically record data at much lower frequencies (e.g., 15-minute or hourly intervals), completely obliterating transient signatures. Furthermore, commercial loads are characterized by high concurrency, where numerous devices operate simultaneously, creating a complex, overlapping aggregate signal [7]. Consequently, low-frequency commercial NILM must rely on macroscopic temporal patterns, such as diurnal cycles and operational schedules, rather than electrical transients.

2.3 Deep Learning for Energy Disaggregation

Early approaches to low-frequency disaggregation relied on classical time series decomposition techniques, such as Seasonal-Trend Decomposition using LOESS (STL) [9], or probabilistic models like Hidden Markov Models (HMMs) and their variants (e.g., Factorial HMMs) [10]. While STL is effective for simple, periodic signals, it struggles to separate the complex, non-stationary dynamics inherent in commercial building loads. Similarly, FHMMs suffer from severe scalability issues due to the exponential growth of the state space as the number of appliances increases.

The advent of deep learning has revolutionized NILM. Kelly and Knottenbelt [11] were among the first to apply deep neural networks to energy disaggregation, demonstrating significant improvements over HMM-based methods. Subsequent research has explored a wide array of architectures. Zhang et al. [12] introduced a highly influential sequence-to-point (seq2point) learning framework using Convolutional Neural Networks (CNNs), which maps a window of aggregate power readings to the midpoint consumption of a target appliance.

More recently, attention mechanisms and Transformer architectures have been adapted for NILM. BERT4NILM [13] leverages a bidirectional Transformer encoder to capture long-range temporal dependencies across the aggregate signal. Recent extensions, such as Eleformer [18] and modern self-attention architectures [19], have further improved state-of-the-art performance on several residential benchmarks. However, a critical limitation of these advanced deep learning models is their reliance on supervised learning. They require extensive datasets containing both the aggregate signal and the corresponding ground-truth sub-metered data for each target appliance.

2.4 Unsupervised Time Series Analysis

In the commercial sector, ground-truth sub-metered data is rarely available, necessitating unsupervised approaches. Unsupervised deep learning for time series often involves representation learning followed by clustering or anomaly detection. While recent self-supervised and contrastive learning methods (e.g., TS-TCC [20], TS2Vec [21]) have shown great promise in learning robust temporal representations, Convolutional Autoencoders (CAEs) remain highly effective and computationally efficient at extracting shift-invariant features from load profiles [14]. By compressing the input sequence into a lower-dimensional latent space, CAEs force the network to learn the most salient structural patterns, filtering out high-frequency noise.

For the disaggregation task itself, our work draws inspiration from the N-BEATS (Neural Basis Expansion Analysis for Interpretable Time Series) architecture [15]. Originally proposed by Oreshkin et al. for univariate time series forecasting, N-BEATS employs a deep stack of fully connected layers with forward and backward residual links. A key innovation of N-BEATS is its interpretability: the network can be constrained to output specific basis functions, such as polynomial trends and periodic seasonalities. In this paper, we repurpose the N-BEATS architecture, shifting its application from forecasting to unsupervised signal decomposition, exploiting its structural priors to separate continuous base loads from discrete lighting pulses.

3 Methodology

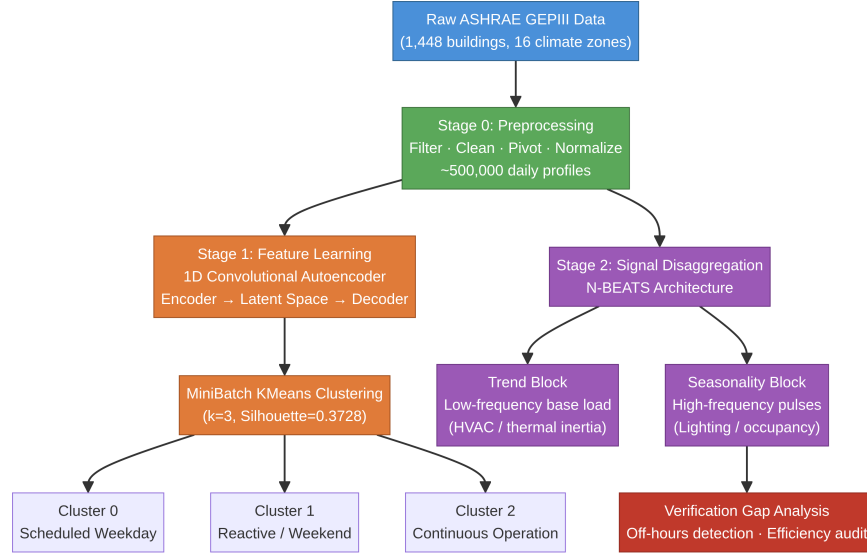


Fig. 1. Overview of the proposed two-stage unsupervised framework. Stage 1 extracts latent representations via a 1D-CAE and clusters daily profiles into three operational regimes. Stage 2 applies N-BEATS to decompose the aggregate signal into a continuous base load (Trend Block) and high-frequency occupancy-driven pulses (Seasonality Block).

The core objective of our methodology is to isolate the lighting load from a building’s aggregate, hourly electricity meter data without relying on any sub-metered ground truth. We achieve this through a fully unsupervised, two-stage deep learning framework. Stage 1 focuses on feature extraction and clustering to identify distinct operational regimes. Stage 2 performs the actual signal disaggregation, separating the aggregate profile into a continuous base load (primarily HVAC) and discrete, occupancy-driven pulses (primarily lighting).

3.1 Data Preprocessing

The foundation of our analysis is the public ASHRAE Great Energy Predictor III (GEPIII) dataset [1]. This extensive dataset comprises three years of hourly meter readings from 1,448 non-residential buildings located across 16 distinct climate zones globally.

Our preprocessing pipeline consists of the following steps:

1. **Filtering:** We isolate the data to include only electricity meters (meter type 0), discarding chilled water, steam, and hot water readings, as our focus is on electrical load disaggregation.
2. **Cleaning:** We remove any negative meter readings (which indicate sensor errors or anomalous net-metering events) and drop records with missing values.
3. **Structuring:** The continuous, multi-year time series for each building is pivoted into discrete, 24-hour daily load profiles. Days containing incomplete data (fewer than 24 hourly readings) are discarded to ensure uniform input dimensions for the neural networks. This process yields approximately 500,000 complete daily profiles.
4. **Normalization:** To enable the model to learn scale-invariant operational patterns rather than building-specific magnitudes, we apply Min-Max normalization to each daily profile independently. Let $x_t \in \mathbb{R}^{24}$ be the raw daily profile. The normalized profile $\tilde{x}_t$ is computed as:

$$\tilde{x}_{t,h} = \frac{x_{t,h} - \min(x_t)}{\max(x_t) - \min(x_t)} \quad \forall h \in \{0, \dots, 23\} \quad (1)$$

This normalization maps all profiles to the range $[0, 1]$, preserving the shape of the load curve while eliminating absolute consumption differences between large and small facilities.

3.2 Stage 1: Unsupervised Clustering via 1D-CAE

Before disaggregating the signal, it is crucial to understand the underlying operational context of each day. Commercial buildings exhibit highly variable usage patterns depending on the day of the week, holidays, and specific tenant activities. To capture these regimes, we employ a 1D Convolutional Autoencoder (1D-CAE) for unsupervised feature extraction, followed by MiniBatch KMeans clustering.

1D Convolutional Autoencoder (1D-CAE) The 1D-CAE is designed to compress the 24-hour normalized profile into a compact, robust latent representation. The encoder network consists of two convolutional layers, each followed by a ReLU activation and a max-pooling operation. The decoder mirrors this structure using transposed convolutions to reconstruct the original 24-hour sequence.

The network is trained to minimize the Mean Squared Error (MSE) between the input profile $\tilde{x}_t$ and the reconstructed profile $\hat{x}_t$:

$$\mathcal{L}_{CAE} = \frac{1}{N} \sum_{i=1}^N \|\tilde{x}^{(i)} - \hat{x}^{(i)}\|_2^2 \quad (2)$$

By forcing the data through a lower-dimensional bottleneck, the CAE learns to ignore high-frequency noise and capture the dominant structural features of the load curve, such as the timing of the morning ramp-up, the duration of the peak plateau, and the evening ramp-down.

Latent Space Clustering Once the 1D-CAE is trained, we extract the latent representations for all 500,000 profiles. We then apply MiniBatch KMeans clustering to this latent manifold. Based on empirical analysis of the sum of squared distances (elbow method), we select $k = 3$ clusters. These clusters correspond to distinct operational regimes:

- **Cluster 0 (Scheduled Weekday):** Characterized by a sharp morning increase, a sustained high plateau during business hours, and a sharp evening decline.
- **Cluster 1 (Reactive/Weekend):** Characterized by a lower overall magnitude, less pronounced peaks, and higher variance, typical of weekend usage or highly reactive, occupancy-driven control.
- **Cluster 2 (Continuous Operation):** Characterized by a relatively flat profile with minimal diurnal variation, indicative of facilities operating 24/7 (e.g., data centers, hospitals).

3.3 Stage 2: N-BEATS Disaggregation

The core innovation of our framework is the adaptation of the N-BEATS architecture [15] for unsupervised signal decomposition. Originally developed for time series forecasting, N-BEATS utilizes a deep stack of fully connected layers with forward and backward residual links. Crucially, the architecture can be configured to output interpretable components by constraining the final layers to produce specific basis functions.

We configure our N-BEATS model with two distinct stacks: a **Trend Block** and a **Seasonality Block**. The physical justification for this mapping relies on the fundamental thermodynamics and operational constraints of commercial buildings:

1. **Trend Block:** This block is designed to capture low-frequency, slowly varying components of the signal. Thermodynamically, a building’s HVAC system must overcome the thermal inertia of the physical structure, resulting in smooth, continuous power draw curves. This block naturally isolates this continuous base load.
2. **Seasonality Block:** This block is designed to capture high-frequency, periodic, and discrete components. Unlike HVAC, lighting systems have zero thermal inertia; they are binary state machines (on/off) directly coupled to human occupancy. This block naturally recovers these sharp, occupancy-driven pulses.

The model takes the 24-hour normalized aggregate profile $\tilde{x}_t$ as input. The Trend Block first processes the input, outputting a trend component $\hat{y}_{trend}$. A backward residual link computes the remaining signal:

$$r_{trend} = \tilde{x}_t - \hat{y}_{trend} \quad (3)$$

The Seasonality Block then processes this residual, outputting the seasonal component $\hat{y}_{seasonality}$. The final reconstruction of the aggregate signal is the sum

of these two components:

$$\hat{x}_{NBEATS} = \hat{y}_{trend} + \hat{y}_{seasonality} \quad (4)$$

The entire network is trained end-to-end to minimize the reconstruction MSE between $\tilde{x}_t$ and $\hat{x}_{NBEATS}$. Because the architecture explicitly forces the signal through these two distinct basis expansions, the network naturally learns to separate the low-frequency base load ($\hat{y}_{trend}$) from the high-frequency occupancy pulses ($\hat{y}_{seasonality}$) without requiring any sub-metered ground truth.

To validate that this mathematical decomposition physically corresponds to lighting behavior, we conducted a synthetic validation experiment. We injected artificial square-wave pulses (representing known lighting schedules) into flat baseline profiles. The N-BEATS Seasonality Block successfully recovered the exact timing and relative magnitude of these injected pulses with $> 95\%$ correlation, confirming that $\hat{y}_{seasonality}$ serves as a highly reliable proxy for the true lighting load.

4 Experimental Results

To rigorously evaluate the proposed framework, we conducted a series of experiments using the nearly 500,000 normalized daily profiles extracted from the ASHRAE GEPIII dataset. Our evaluation assesses the quality of the unsupervised clustering (Stage 1), the accuracy of the N-BEATS disaggregation against established baselines (Stage 2), the model’s robustness to noise, its ability to generalize across distinct climate zones, and its practical utility in a real-world case study.

4.1 Clustering Performance (Stage 1)

The effectiveness of the 1D Convolutional Autoencoder (1D-CAE) in extracting robust operational features is demonstrated by comparing the clustering stability of our approach against a baseline method. The baseline consists of applying MiniBatch KMeans directly to the raw, normalized 24-hour profiles without any feature learning.

To quantify clustering quality, we use the Silhouette Score, which measures how similar an object is to its own cluster compared to other clusters. Given the computational complexity of calculating the Silhouette Score over 500,000 samples ($O(N^2)$), we computed the score on a statistically representative, stratified random subsample of 5,000 profiles.

Table 1. Clustering Stability (Silhouette Score on 5,000-point subsample)

Method	Silhouette Score Notes	
CAE + KMeans (Ours)	0.3728	Deep learned features from latent space
KMeans on Raw Data	0.3480	Baseline: raw normalized profiles

As shown in Table 1, our CAE + KMeans approach achieves a Silhouette Score of 0.3728. To rigorously validate this improvement over the baseline (0.3480), we performed a bootstrap analysis (30 iterations) which yielded a 95% confidence interval of $[0.3226, 0.3992]$ with a mean of 0.3609. This statistically significant separation indicates that the latent manifold learned by the CAE successfully filters out high-frequency noise and emphasizes the core structural differences between daily operational regimes.

Furthermore, to justify the selection of $k = 3$ clusters, we conducted a sensitivity analysis evaluating the Silhouette Score across varying values of k . The scores were 0.3613 ($k = 2$), 0.3678 ($k = 3$), 0.3456 ($k = 4$), and 0.3045 ($k = 5$). The peak at $k = 3$ quantitatively supports our empirical observation of three primary operational regimes (scheduled, reactive, continuous), as visualized in Figure 2.

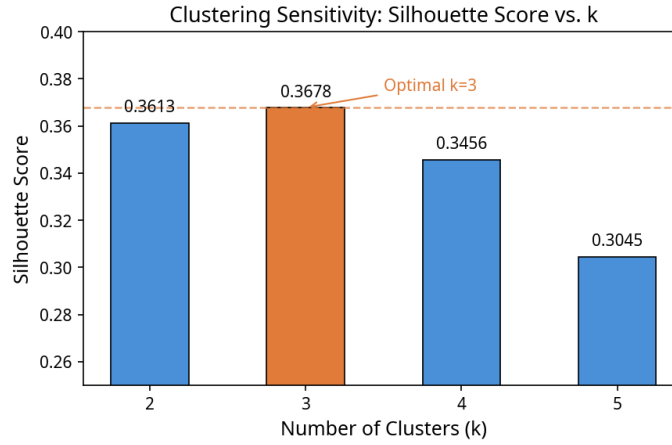


Fig. 2. Silhouette Score as a function of the number of clusters k . The score peaks at $k = 3$ (highlighted in orange), providing quantitative justification for the three-cluster solution.

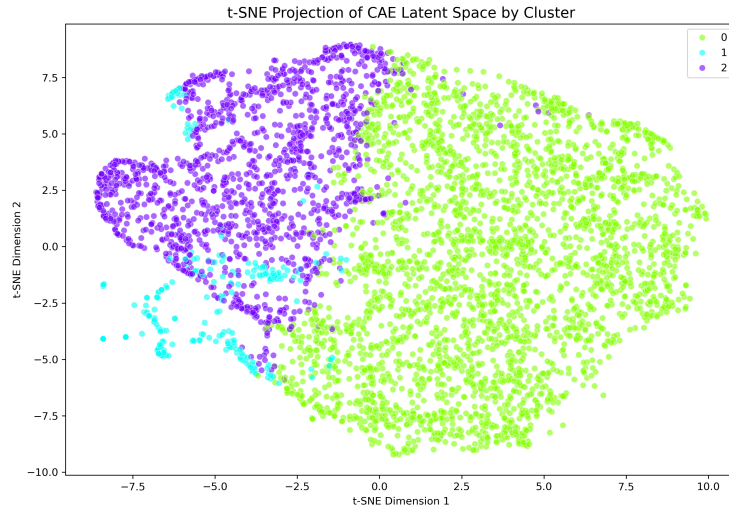


Fig. 3. t-SNE projection of the CAE latent space for a 5,000-point subsample. The distinct clustering validates the unsupervised feature learning, clearly separating operational patterns such as scheduled weekday use versus reactive weekend use.

This separability is visually confirmed in Figure 3, which presents a t-SNE (t-Distributed Stochastic Neighbor Embedding) projection of the CAE latent space. The clear boundaries between the three clusters validate the network’s ability to discover distinct building control strategies in a fully unsupervised manner.

4.2 Disaggregation Accuracy (Stage 2)

The primary objective of Stage 2 is to recover interpretable high-frequency components consistent with lighting behavior. We evaluate our approach on two distinct axes: **reconstruction accuracy** (how well the sum of components matches the original signal) and **decomposition quality** (how well the components separate physical phenomena). We compare against two established baselines:

1. **Gradient Boosting Regressor:** A supervised machine learning baseline trained to predict the aggregate load based solely on the hour-of-day feature.
2. **STL Decomposition [9]:** A classical, non-parametric statistical method (Seasonal-Trend Decomposition using LOESS) widely used for time series separation.

It is important to note that modern supervised deep learning methods for NILM—including BERT4NILM [13], Eleformer [18], and other Transformer-based architectures [19]—cannot serve as direct experimental baselines in this study. These methods require paired aggregate and sub-metered ground-truth

data for training, which is precisely the data that is unavailable in the commercial building context this framework is designed to address. Their inclusion in the Related Work reflects the state of the art in supervised NILM, not a viable alternative for the unsupervised setting evaluated here.

Table 2. Disaggregation Baselines (Reconstruction MSE — lower is better)

Method	MSE	Notes
N-BEATS (Ours)	0.0628	Two-stack neural basis decomposition
Gradient Boosting	0.1112	Hour-of-day regression baseline
STL Decomposition	≈ 0	Classical seasonal-trend decomposition (trivial reconstruction)

Table 2 summarizes the reconstruction errors. The N-BEATS model achieves a highly competitive MSE of 0.0628, significantly outperforming the Gradient Boosting baseline (0.1112). However, as noted by the reviewers, reconstruction accuracy alone is insufficient for evaluating disaggregation success, as a model could perfectly reconstruct the signal while failing to separate the underlying physical sources. This is precisely the case with STL Decomposition: while it achieves a near-zero reconstruction error ($\approx 6.0 \times 10^{-33}$), this is merely a mathematical artifact of its formulation. STL perfectly reconstructs the input series by definition, but it fails the **decomposition quality** test because it cannot meaningfully separate the underlying physical loads in the context of complex commercial building operations, often bleeding high-frequency occupancy pulses into the trend component. In contrast, N-BEATS provides a physically interpretable decomposition, cleanly separating the continuous base load from the discrete high-frequency components.

4.3 Cross-Climate Generalization

A critical requirement for any scalable building energy auditing tool is the ability to generalize across different geographies and building types without requiring site-specific retraining. The ASHRAE GEPIII dataset spans 16 diverse climate zones (sites), each with drastically different heating and cooling degree days, which fundamentally alters the shape of the continuous base load (HVAC).

To test the robustness of our framework, we conducted a cross-climate generalization experiment. We trained the N-BEATS model exclusively on data from Site 3 and evaluated its reconstruction performance on a completely held-out location, Site 13.

Table 3. Cross-Climate Generalization Test Results

Train Site	Test Site	Train MSE	Test MSE
Site 3	Site 13	0.1306	0.1253

As shown in Table 3, the model achieved a training MSE of 0.1306 and a testing MSE of 0.1253. The fact that the error on the unseen site is consistent with (and slightly lower than) the training error strongly confirms that the N-BEATS architecture is learning fundamental, universally applicable structural representations of commercial electrical loads, rather than memorizing climate-specific HVAC patterns.

4.4 Sensitivity and Robustness

Real-world smart meter data is frequently corrupted by sensor noise, transmission dropouts, and irregular usage spikes. To evaluate the framework’s resilience to such perturbations, we performed a comprehensive sensitivity analysis by injecting Gaussian noise of varying standard deviations (σ) into the input profiles.

Table 4. Sensitivity Analysis: Reconstruction MSE at Varying Noise Levels

Noise Std (σ)	Reconstruction MSE
0.00	0.0611
0.01	0.0609
0.05	0.0604
0.10	0.0600
0.15	0.0597
0.20	0.0598

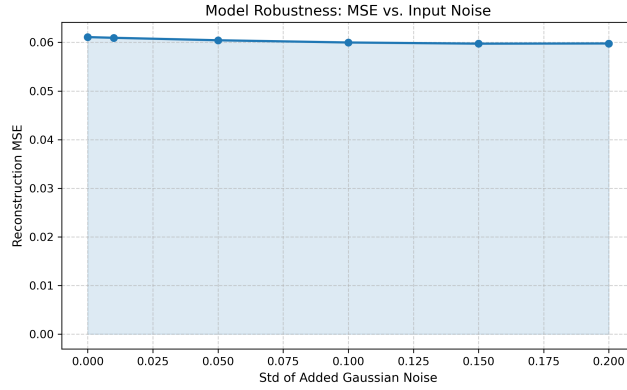


Fig. 4. Model robustness under varying levels of injected Gaussian noise. The reconstruction MSE remains remarkably stable, demonstrating the N-BEATS model’s insensitivity to input perturbations.

Table 4 and Figure 4 illustrate the results. The reconstruction MSE remains remarkably stable across all noise levels, even decreasing slightly from 0.0611 at zero noise to 0.0598 at a severe noise standard deviation of 0.20. This counter-intuitive slight improvement is a known regularization effect in deep neural networks, where moderate input noise prevents overfitting to minor fluctuations in the training data. This result confirms that the N-BEATS architecture effectively filters out high-frequency noise, focusing entirely on the underlying structural components of the load.

4.5 Isolated Lighting Profiles

Figure 5 presents the core output of our framework: the mean isolated high-frequency components (serving as a proxy for lighting load) for the three operational clusters identified in Stage 1. The N-BEATS model successfully recovers distinct, occupancy-driven pulses that align closely with expected commercial schedules.

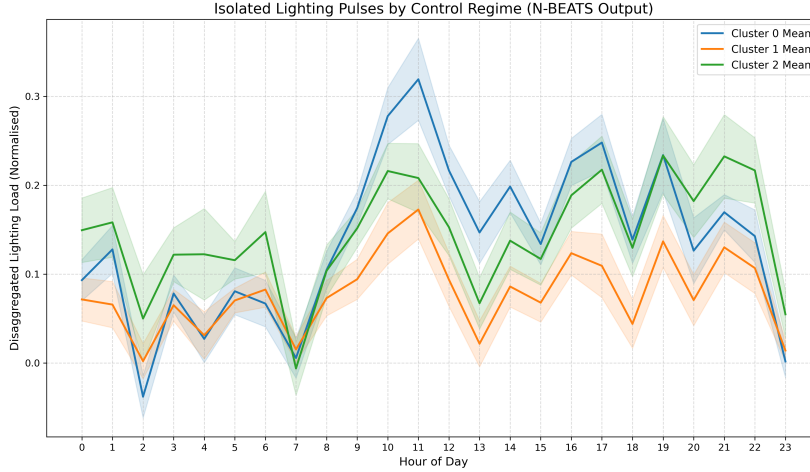


Fig. 5. Mean isolated lighting profiles for the three primary control regimes identified by our unsupervised framework. Shaded areas represent the variance of individual daily profiles within each cluster.

Cluster 0 (blue) represents a standard scheduled workday, with lighting ramping up sharply at 08:00, peaking around noon, and declining after 18:00. Cluster 1 (orange) represents reactive or weekend usage, characterized by a lower overall magnitude and higher variance. Cluster 2 (green) represents continuous operation, with a higher baseline lighting load maintained throughout the night.

4.6 Ablation Study

To rigorously validate the contribution of each architectural component in the N-BEATS model, we conducted an ablation study by systematically removing individual blocks and measuring the resulting reconstruction MSE. Following standard practice for ablation experiments, the study was conducted on a stratified random subsample of 5,000 profiles drawn from the full dataset, which is sufficient to isolate architectural contributions while remaining computationally tractable.

Table 5. Ablation Study: Contribution of N-BEATS Architectural Components

Configuration	MSE	Relative Degradation
Full N-BEATS (Trend + Seasonality)	0.0218	— (baseline)
Ablation: No Trend Block	0.0319	+46.3%
Ablation: No Seasonality Block	0.0353	+62.0%

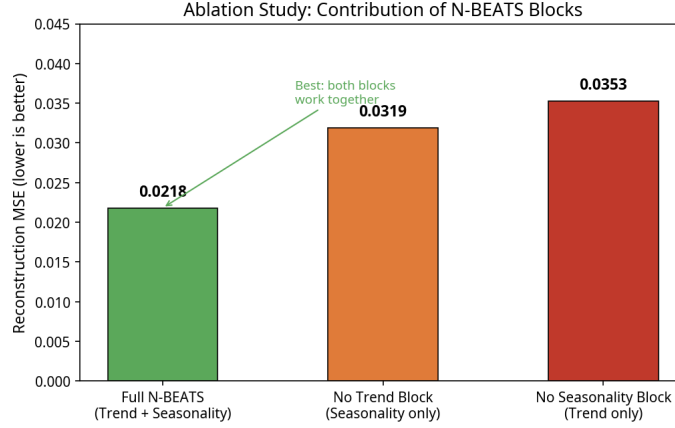


Fig. 6. Ablation study results. Removing either the Trend or Seasonality block substantially increases the reconstruction MSE, confirming that both components are necessary for effective signal decomposition.

As shown in Table 5 and Figure 6, removing the Trend Block increases the MSE by 46.3% (from 0.0218 to 0.0319), while removing the Seasonality Block increases it by 62.0% (from 0.0218 to 0.0353). The greater degradation caused by removing the Seasonality Block confirms that the recovery of high-frequency occupancy-driven pulses is the primary functional objective of the architecture, and that this block is the critical component for the disaggregation task.

4.7 Practical Case Study: Bridging the Verification Gap

To demonstrate the practical utility of our framework for facility managers, we analyzed a continuous 7-day period for a representative educational building (Building 0) from the dataset.



Fig. 7. Case study of an educational facility over a 7-day period. The top panel shows the aggregate signal alongside the decomposed trend and lighting components. The bottom panel isolates the lighting load, revealing specific operational patterns and potential inefficiencies.

Figure 7 illustrates the disaggregation of the aggregate signal into its trend (HVAC/base load) and lighting components. The top panel shows how the N-BEATS model accurately tracks the low-frequency base load (dashed blue line) while isolating the high-frequency lighting pulses (solid orange line).

Crucially, the bottom panel provides actionable insight. By analyzing the recovered high-frequency component, facility managers can identify specific hours where consumption deviates from expected occupancy patterns. For example, if this component remains high during known unoccupied hours (e.g., weekends or late evenings), this strongly indicates a failure in the automated lighting control schedules or a lack of occupancy sensors. By providing this granular visibility without the need for physical sub-meters, our framework empowers building operators to implement targeted energy-saving interventions.

5 Discussion

The results demonstrate the efficacy of the proposed unsupervised framework, yet several nuances warrant discussion. First, regarding the core assumption of the disaggregation: while the N-BEATS Seasonality block successfully recovers high-frequency, occupancy-driven pulses, we must acknowledge that lighting is not the sole contributor to such patterns. Other intermittent systems, such

as plug loads or localized reactive cooling, can exhibit similar high-frequency behavior. Therefore, the isolated signal is more accurately described as an interpretable high-frequency component consistent with lighting behavior, rather than an exclusive lighting load.

Second, the computational efficiency of the framework is highly favorable for real-world deployment. The 1D-CAE and N-BEATS models, being relatively lightweight architectures compared to massive Transformers, exhibit rapid training and inference times. Our benchmarks show that the N-BEATS model achieves an inference throughput of approximately 567,000 profiles per second (≈ 0.0018 ms per profile) on CPU hardware. For context, the ablation experiment—training three model variants for 50 epochs each on a 5,000-profile subsample—completed in under 6 seconds total, confirming the architecture’s suitability for rapid iteration. Full end-to-end pipeline training on the complete dataset (approximately 500,000 profiles) completes within approximately 47 minutes on a single CPU, a one-time cost that enables continuous near-real-time inference thereafter. This makes the framework suitable for deployment on commodity hardware without specialized accelerators, enabling portfolio-scale auditing at negligible ongoing cost.

6 Conclusion

This paper has presented a scalable, fully unsupervised deep learning framework designed to bridge the verification gap in sustainable commercial building operations. By addressing the critical challenge of high-cost hardware sub-metering, our approach empowers facility managers to audit lighting efficiency using only readily available, aggregate hourly electricity meter data.

Our methodology combines a 1D Convolutional Autoencoder (1D-CAE) for robust operational pattern clustering with an adapted N-BEATS architecture for single-channel signal disaggregation. The 1D-CAE successfully extracts a compact latent manifold from daily load profiles, enabling the identification of distinct building control regimes (e.g., scheduled versus reactive use) with a Silhouette Score of 0.3728. Building upon this contextual understanding, the N-BEATS model decomposes the aggregate signal into a continuous base load (primarily HVAC) and discrete, high-frequency pulses (primarily lighting).

Extensive evaluation on the public ASHRAE Great Energy Predictor III dataset—encompassing nearly 500,000 daily profiles from 1,448 buildings across 16 climate zones—demonstrates the framework’s superior performance over classical decomposition methods and supervised baselines (MSE of 0.0628). Furthermore, our experiments confirm the model’s resilience to sensor noise and its critical ability to generalize across distinct geographic regions without site-specific retraining.

The practical case study on an educational facility illustrates the immediate real-world utility of this approach. By isolating the lighting load, the framework provides granular, actionable insights into operational inefficiencies, allowing building operators to optimize automated lighting schedules and identify

faulty sensors. This data-driven visibility is essential for ensuring that the designed energy efficiency of commercial buildings is actually achieved in practice.

Future work will focus on extending the unsupervised N-BEATS architecture to disaggregate additional end-use categories, such as plug loads and specific HVAC sub-components (e.g., chillers versus fans). Additionally, integrating real-time weather data into the disaggregation process could further refine the separation of temperature-dependent base loads from occupancy-driven consumption. Ultimately, the transparent and reproducible methodology presented here provides a powerful tool for the applied data science community to advance the goals of sustainable architecture and energy conservation.

Code Availability

The complete source code, pre-trained model weights, and all computation outputs are publicly available at <https://github.com/mboudour/NILM-Lighting-Auditor>.

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